

A Fuzzy-IoT Irrigation Framework with LoRa-Based Connectivity for Climate-Resilient Agriculture in Semi-Arid Regions

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Highlights

- Developing a fuzzy-IoT irrigation system for semi-arid agriculture.
- Integrating multi-sensor data for adaptive irrigation decision-making.
- Enabling LoRa-based long-range communication without continuous internet.
- Reducing irrigation water use by 26% in field deployment.
- Enhancing sustainability compared with conventional and existing fuzzy systems.

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Abstract

In arid and semi-arid regions, water scarcity is one of the most significant challenges in agriculture. Efficient water management is important where precipitation is low, and groundwater resources are also depleting. Most existing systems rely on a single sensor input, require continuous internet connectivity, and offer limited adaptability to changing field conditions. This research presents a Fuzzy-IoT irrigation system, implemented for the GC-2 variety of castor in Bhuj, India. The system combines data from multiple sensors, including soil moisture, rainfall, temperature, and relative humidity. The system consists of a sensor node and a collector node. The sensor node measures field parameters and sends them to the collector node. The collector node performs local fuzzy-logic decision-making to control the pump without an internet connection and uploads data to the ThingSpeak cloud via an ESP-based module for monitoring. In this system, data are applied to a fuzzy inference framework to enhance irrigation scheduling and improve decision-making in semi-arid regions. Moreover, to ensure practical usability in rural areas, the framework also utilizes LoRa, which provides infrastructure-less communication and long-range operation without continuous internet access. Field evaluation revealed that the system reduced the total irrigation water requirement to 337.91 m³/acre, achieving 26% water savings. Specifically, this performance in water saving is higher than the other state-of-the-art fuzzy-based irrigation approaches. Overall, the results demonstrate that fuzzy IoT irrigation has delivered sustainable water savings compared with traditional irrigation practices in semi-arid agricultural regions.

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1. Introduction

India's economy depends heavily on agriculture. Agriculture in India provides 55% of jobs directly and even more indirectly [1]. While agriculture's contribution to India's GDP has decreased over time as other industries have grown, it continues to play an important role. Agriculture exports from India achieved a record high of USD 50 billion in the fiscal year 2020-21 [2]. Agriculture in India is continuously evolving, and the use of artificial intelligence, the Internet of Things, Drones,

Computer Vision, and Robotics can be observed everywhere [3]. India faces serious difficulty in managing irrigation water, an essential resource for agriculture. Water demand for Irrigation has increased due to population growth and climate change. Punjab, Rajasthan, and Tamil Nadu overexploited groundwater by 76%, 63%, and 40%, respectively, for irrigation [4]. The use of irrigation techniques (drip and sprinkler irrigation) [5] is popular for reducing irrigation water requirements. Meeting the water requirement for

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Irrigation is a major problem in India. North Gujarat and Bhuj (Kutch) are part of Gujarat (Indian state) and face water scarcity due to their arid and semi-arid climate. Bhuj and other parts of the North Gujarat region are experiencing a lack of rainfall, higher temperatures, and higher evapotranspiration, along with lower humidity. The vegetative cover in this region is also very low. In the Run of Kutch, average annual rainfall is around 14 to 38 cm, which is the lowest in Gujarat [6]. Drought is observed on average once in every three years in Bhuj (Kutch), a reality deeply ingrained in the lives of its residents. However, the severity of each drought varies from year to year. The year 2019 has been particularly harsh, marking the worst conditions since the drought of 1985-88 [7]. These situations highlight the desperate need for sustainable water management strategies in Bhuj (Kutch), Gujarat.

Agricultural automation [8] is a fundamental technique that reduces labor dependency without

compromising productivity and improves large-scale farming. Agricultural automation, precision farming [9], data analytics, farm management software [10], advances in biotechnology and genetically modified crops [11] are the main approaches witnessed in recent years. To maximize crop yields and operational efficiency, smart irrigation systems are heavily used in Precision farming [12-14]. They are helping farmers by providing the water crops need. Due to the availability of high-precision, inexpensive sensors, farmers are shifting towards smart irrigation systems to monitor and exploit parameters such as soil moisture, rainfall, and solar radiation [15,16].

On-site and remote sensing techniques are widely employed to gather various parameters in real time, which helps in making irrigation decisions. Farmers can save water and fertilizers by choosing precise amounts through smart irrigation practices [17].



Fig. 1. Geographic Context of the Study Area: Gujarat, India.

Modern Internet of Things (IoT) technologies provide opportunities to address challenges in agriculture, particularly by enhancing sustainability, improving crop yields, and increasing cost-efficiency [18]. Smart irrigation systems [19,20] support precision agriculture practices by utilizing real-time data from soil moisture sensors, weather inputs, and crop-specific requirements to optimize water delivery [21-23]. Additionally, lifecycle assessment (LCA) methodologies are increasingly applied to evaluate the environmental and health impacts of crop production in India [24]. Machine learning is playing an important role in agricultural applications. It helps with air pollution and rainfall forecasting [25,26], groundwater level prediction [27], and modeling the relationships among various

parameters [28]. The integration of IoT with machine learning can help optimize resources, reduce environmental impact, and enable data-driven agricultural decision-making [29]. Crop selection can also be improved by considering multiple factors, such as production output, water demand, emissions, and agricultural residues. Smart irrigation systems incorporate IoT devices and environmental sensors to continuously monitor field conditions and address the problems of traditional irrigation schedules. This real-time monitoring enables predictive and adaptive control of irrigation processes [30,31].

Often, additional infrastructure is required on the farm to deploy electronic devices and sensors effectively [32]. Generally, wired connections are not preferred due

to cost, and cables make them vulnerable to damage from animals, plant roots, and the regular movement of tractors and other machinery [33]. For these reasons, wireless communication technologies are commonly adopted. LoRa and ZigBee are popular wireless technologies from a power-efficiency perspective [34]. LoRa provides long-range communication across a large farm area without compromising battery life [35]. Transmission levels are adjusted via software to balance communication reliability with energy efficiency. It also meets local regulatory limits for many real-life applications. Raspberry Pi is highly efficient in terms of CPU performance. However, it requires cooling to prevent overheating during prolonged computational use. The ARM-based STM32 microcontroller [36] requires less power, is low-cost, and operates at a high frequency (84 MHz). Raspberry Pi's quad-core processor operates at 1.5 GHz, which makes it more powerful than the STM32 microcontroller. It may be excessive for simple sensing and control tasks. This research aims to implement a Fuzzy-IoT irrigation system for Bhuj, with a main focus on water saving. This system addresses the challenges of Irrigation in rural areas, where farms are often situated 2-30 kilometers away from urban areas with no or limited internet connectivity. For implementation, it ensures that field components can operate autonomously without using internet

connectivity. The proposed system uses LoRa wireless communication and an STM32 ARM microcontroller.

The key contributions of this study are summarized as follows:

- A novel approach that integrates data from multiple sensors (humidity, temperature, soil moisture) to optimize irrigation management and improve decision-making using Fuzzy-logic.
- Infrastructure-less connectivity using LoRa, enabling long-range communication and reliable operation without the need for continuous internet access in the field.
- The proposed system addresses water scarcity challenges more effectively than existing fuzzy-IoT approaches and traditional irrigation methods.
- Demonstrated water savings of 26%, outperforming state-of-the-art fuzzy-IoT systems.

The remainder of this paper is structured as follows: Section 2 details the site conditions, smart irrigation methods, and the evaluation parameters considered in this study. Section 3 presents the methodology, including the experimental setup for the proposed Fuzzy-IoT Irrigation System. Section 4 describes the implementation of the proposed model. Finally, Section 5 presents and discusses the results obtained from the experiments.

Average Yearly Rainfall(mm) in Gujarat(1990-2019)

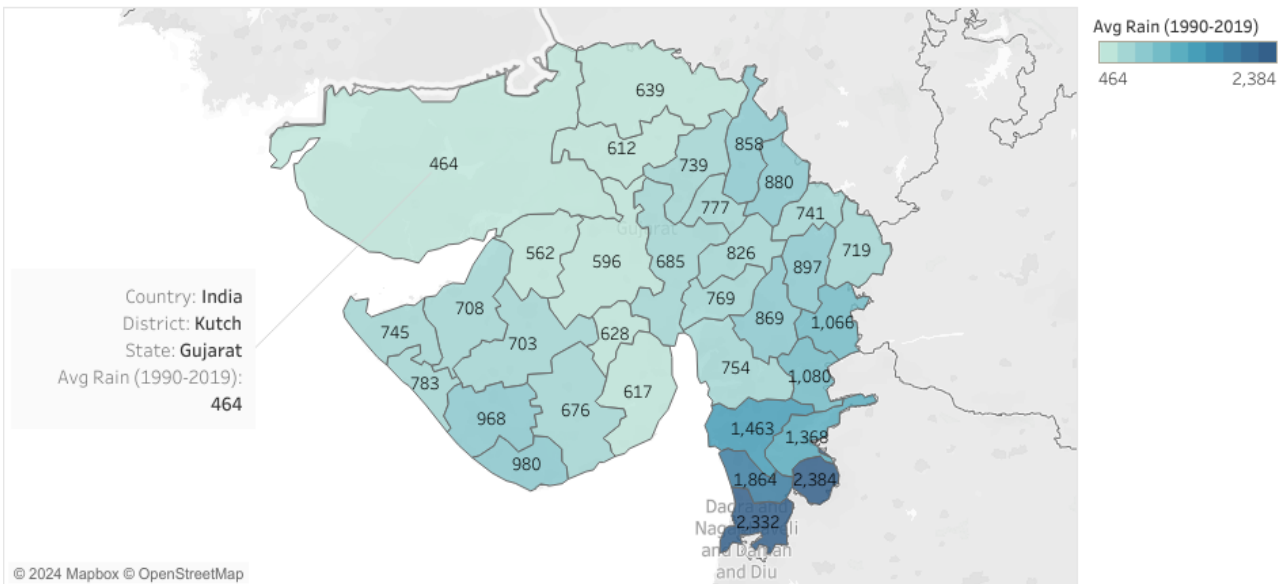


Fig. 2. Average Annual Rainfall by District in Gujarat.

2. Methodology

The experiment implements an irrigation system in the Bhuj(Kutch) district of Gujarat, India. The climate in Bhuj is semi-arid, which has a significant impact on agricultural farming methods. Fig. 1 shows a location map of the study area in India. The average annual rainfall

from 1999 to 2019 for the Bhuj (Kutch) region is shown in Fig. 2.

2.1.Agricultural Field Study: Site Conditions

Bhuj experiences a hot summer, a cold winter, and below-average rainfall in Gujarat. Summer temperatures range from 35 °C to 45 °C, and winter temperatures range from 8 °C to 25 °C. This part of Gujarat witnesses an

average rainfall of 300-400 mm annually. Water conservation is important for farming due to the very high evaporation and low humidity [37]. The soil is tested at a depth of 15-30 cm, and the following observations are recorded: pH (9.67), bulk density (2.61%), particle density (4.18%), porosity (39.63%), and water holding capacity (57.72%). Short-duration castor seed varieties, such as Jyothi, Kranti, and GC-2, are popular in this region for their high yields. In this research work, GC-2's well-established robustness and high yield under the local climate were chosen. The soil mulching technique was used to reduce moisture loss and improve water retention [38].

2.2. Smart Irrigation Methods

Smart irrigation systems can be categorized based on characteristics such as the decision-making criteria used to irrigate, the technological implementation, and the method of water supply. These characteristics are often mutual to develop effective, dynamic, situation-based solutions. For decision-making, systems may use different parameters, such as:

- Soil-based parameters: moisture level, suction pressure.
- Climate factors: rain, wind speed, humidity, temperature.
- Plant-based features: the amount of water in the leaves, crop growth level, yield target, and precise water requirements.
- Soil appearances: texture, porosity, and water-holding capacity.

Smart irrigation systems can be divided into sensor-based, cloud-based, IoT, and machine learning-based systems [39-41].

Additional solutions incorporate evapotranspiration models, soil moisture forecasting, and data-driven machine learning models to provide efficient irrigation [42]. Today, irrigation systems emphasize energy efficiency, low-power communication technologies such as LoRa [43,44], and reduced reliance on manual supervision, especially in rural areas. Moreover, fuzzy logic-based systems demonstrate effectiveness in managing uncertainty and nonlinear conditions by integrating diverse parameters, such as soil moisture, climatic data, and crop-specific water needs [45,46]. These systems are effective in both homogeneous and heterogeneous soil conditions. Also, several smart systems have been presented to specific crop requirements, enhancing precision and yield. For specific crops such as potatoes [43], litchi [45], and sorghum [47], state-of-the-art solutions have been implemented across different farming types and regions to achieve water savings.

2.3. Evaluation Parameters

Smart irrigation systems are designed with key parameters in mind to evaluate their efficiency in water conservation and crop yield improvement. The key parameters are

- The water requirements for irrigation vary among different soils and crops. Water use in irrigation is measured using various techniques and evaluation metrics. After collecting water-use data, it can be displayed in various forms. Water use is commonly represented by the total amount of water distributed over a particular time frame, such as daily, weekly, or monthly, as well as the amount of water consumed per unit area, such as liters per hectare or gallons per acre. Water efficiency is a metric that quantifies the ratio of water use to water demand for certain plants to achieve optimal development. This metric serves as an indicator of the efficiency with which water is used in irrigation. Water-use efficiency measurements, such as agricultural water productivity or irrigation efficiency indices, are commonly used in academic research.

To calculate water requirement for irrigation, a lot of factors need to be evaluated:

- Reference Evapotranspiration (ET_0) is the primary factor in most water requirement estimations. It represents the total water loss from a particular crop due to evaporation and transpiration under given climate conditions.

$$ET_0 = \frac{0.408\Delta(R_n - H_s) + \gamma \frac{900}{T + 273} x_2 (P_s - P_a)}{\Delta + \gamma(1 + 0.34x_2)} \quad (1)$$

where, R_n is net radiation at the crop surface; H_s is soil heat flux density; T is mean daily air temperature at 2m height; P_s is saturation vapour pressure; P_a is actual vapour pressure; Δ is the slope of the vapour pressure curve; and γ is the psychrometric constant.

- Crop water requirement (ET_c): It represents the actual water needs of a crop, accounting for its growth stage and unique attributes.

$$ET_c = K_c \times ET_0 \quad (2)$$

Where K_c is the crop coefficient and ET_0 is the reference evapotranspiration.

- Net irrigation depth (NIR): The amount of water required for Irrigation, taking into account rainfall and soil moisture content.

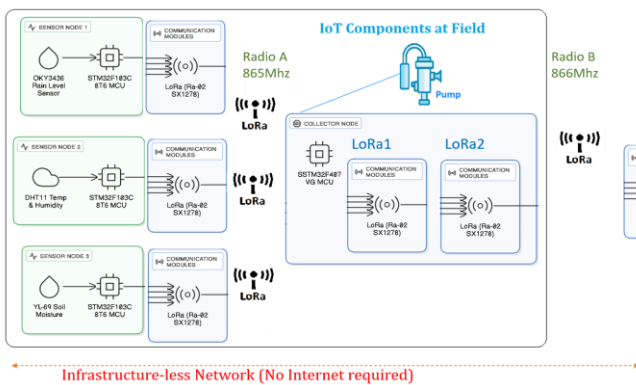
$$NIR = ET_c - P - D \quad (3)$$

Where D and P are deep percolation and effective rainfall, respectively.

- Key parameters, such as flow rate, time, or volume, are incorporated in irrigation using distinct equations. This calculation also includes parameters such as furrow geometry, field slope, infiltration rate, and flood parameters, including water depth and field acreage. These equations are used to frame fuzzy rules for irrigation (Table 2).

- **Energy consumption:** This parameter focuses on the power consumption of pumps or other components that consume substantial energy. Reduced energy usage enables the implementation of more effective watering methods [48].
- **Economic efficiency:** These parameters assess the economic feasibility of the irrigation systems. Additionally, it is recommended to utilize open-source software and low-power sensors and microcontrollers [49].

3. Design and Implementation of the Fuzzy-IoT Irrigation System



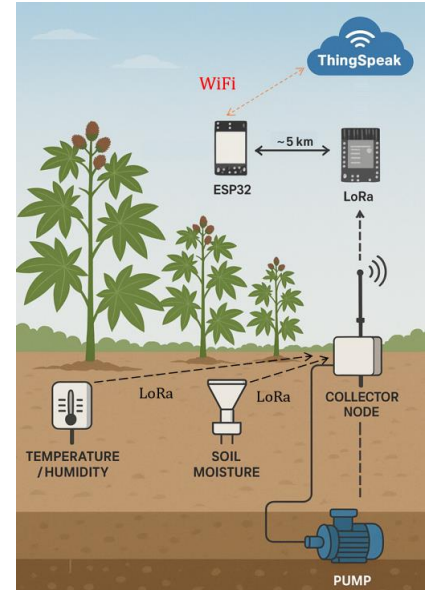
System Architecture of the Fuzzy-IoT Irrigation System

3.1. Sensor Selection and Calibration

The selection of sensors was based on accuracy, power consumption, and compatibility with the STM32 microcontrollers. The system utilizes the following sensors:

- **Soil Moisture Sensor(YL-69):** YL-69 is a low-cost soil moisture sensor widely used in agriculture, gardening, and IoT projects. It helps determine the soil's moisture content. The YL-69 soil moisture sensor is affordable, easy to use, and measures soil moisture using conductivity between the probes. It operates over a 3.3V - 5V range and provides both types of output (Analog and Digital). It can be easily interfaced with Arduino, ESP32, STM32, and other microcontrollers.
- **Digital Relative Humidity and Temperature Sensor (DHT11):** Ambient temperature directly affects the rate of transpiration from crops. Increased temperature results in more water evaporation, increasing the amount of water needed for Irrigation. Increased relative humidity leads to higher

The architecture of the proposed Fuzzy-IoT Irrigation system is presented in Fig. 3. The control framework is built through a sensor integration approach, in which multiple sensor nodes are connected to a single central node unit (Collector node). The sensors interfaced with STM32F103C8T6 are connected to the Collector node, which transmits data over a LoRa transceiver (Ra-02 SX1278). Radio A uses 865.0625 MHz for dedicated communication between field sensor nodes and collector nodes. Radio B uses 866.3000 MHz for communication between the ESP32 node and the Controller node. The ESP32 node is responsible for uploading data to the ThingSpeak server.



Field Deployment of the Fuzzy-IoT System

Fig. 3. Fuzzy-IoT Irrigation System Perspective.

precipitation rates, replenishing groundwater levels and decreasing the demand for crop irrigation from reservoirs. Therefore, it is crucial to monitor the relative humidity (%) to forecast rainfall. It is precalibrated using resistive sensor technology combined with an NTC thermistor to provide accurate relative humidity and ambient temperature readings.

- **Rainwater level sensor (OKY-3436):** OKY-3436 is a rainwater level sensor module. It has been used to detect the presence of rainwater through a conductive sensing plate. When rainwater drops onto the sensor plate, it forms a conductive path across the copper traces. This changes the resistance, which the module converts into an analog voltage signal. The OKY-3436's primary advantage is its affordable, user-friendly rain-detection module.
- **Node MCU (ESP32):** It is used with a Wi-Fi- and Bluetooth-enabled device to communicate with open-source platforms while interfacing with hardware in system design. The ESP32 Wi-Fi + Bluetooth module features a dual-core System on a

Chip (SoC) that integrates a built-in TCP/IP protocol stack, enabling microcontrollers to connect seamlessly to Wi-Fi networks and exchange data over Bluetooth (Classic and BLE). The ESP32 can run standalone applications or delegate network tasks to its dedicated communication processor. Each ESP32 module supports multiple development frameworks (such as the Arduino IDE, ESP-IDF, and MicroPython) rather than relying solely on AT commands. Its robust on-board processing power, integrated memory, and hardware peripherals allow it to interface with sensors, actuators, and other application-specific devices through GPIO, ADC, DAC, I²C, SPI, UART, and PWM with minimal predevelopment effort and runtime overhead. Due to its high on-chip integration, less external circuitry is required, making it suitable for compact PCB designs.

- **Lora Antenna 865 MHz Wi-Fi Transmitter Module:** This device operates on the 865-867 MHz ISM band. It uses LoRa modulation technology (Long Range, chirp spread spectrum). The benefit of using it for long-range, low-power, low-data-rate communication. The module is an ideal antenna for IoT applications.
- **Microcontroller - Sensor Node (STM32F103C8T6):** This MCU belongs to the STM32F1 series, which is based on the ARM Cortex-M3 core. It offers moderate performance, with a maximum CPU frequency of 72 MHz. It has fewer peripherals and features than more advanced series, such as the STM32F4.

- **Microcontroller - Collector Node (STM32F407VG):** This MCU is part of the STM32F4 series and is based on the ARM Cortex-M4 core. It offers higher performance, with a maximum CPU frequency of 168 MHz. It comes with a rich set of peripherals, including USB, Ethernet, CAN, and more, making it suitable for a wider range of applications, including high-performance computing, digital signal processing, and connectivity. Although they are all microcontroller units (MCUs) from the STM32 family of devices made by STMicroelectronics, the STM32F407VG [50,51] and the STM32F103C8T6 are from separate series and have distinct features, pricing points, and power consumption profiles. These are comparable in several aspects as follows:

- **Price:** Generally, the STM32F103C8T6 is cheaper than the STM32F407VG.
- **Power Consumption:** The STM32F103C8T6 typically consumes less power than the STM32F407VG due to its simpler architecture and lower operating frequency.

STM32F407VG would be a better choice for the collector node despite its higher price and potentially higher power consumption. However, if cost and power efficiency are critical factors for our application, the STM32F103C8T6 might be more suitable for the end node, especially for lower-performance requirements. The detailed specifications of Sensors and components are tabulated in Table 1.

Table 1. Detailed specifications of Sensors and components.

Component	Specification
Soil Moisture Sensor(YL69)	Operating voltage (V _{cc}): 3.3V-5V. Analog output voltage: 0-4.2V @ V _{cc} = 5V. Digital output voltage: 0V or 5V @ V _{cc} = 5V. Current consumption: 15 mA.
DHT11 ^a	Humidity measuring range: 20-90%. Temperature measuring range: 0-50 °C. Humidity accuracy: ±5.0%. Temperature accuracy: ±2.0 °C. Response time: <5 s; Input voltage: 5V DC.
Rain water level sensor (OKY-3436)	Operating voltage (V _{cc}): 3.3V-5V. Analog output voltage: 0-4.2V @ V _{cc} = 5V. Digital output voltage: 0V or 5V @ V _{cc} = 5V. Current consumption: 30 mA.
SX1278 ^b	Center frequency: 865 MHz. Supply Voltage: 1.8 - 3.7 V Channel Bandwidths: 7.8 kHz to 500 kHz
Variable Frequency Drive (VFD)	Voltage Rating: 380V to 500V AC. Power Rating: 2 to 5 HP.
TP4056 1A ^c	Safety features: charges a single-cell 3.7V, 1Ah battery.
18650 Battery ^d	Rated at 3.7V DC, 2000 mAh.
Microcontrollers	Main Node: STM32F407VG; End Node: STM32F103C8T.
ESP32 Module	Wi-Fi module; operating voltage: 3.3V or 5V.

^a Digital Relative Humidity and Temperature Sensor Module

^b LoRa Antenna 865 MHz and Transceiver Module

^c 1A Micro USB Lithium-Ion Battery Charger Board

^d Li-ion Power Battery Cell.

3.2. Hardware Setup and Component Interactions

The field setup of proposed Fuzzy-IoT Irrigation System is illustrated in Fig. 3. This IoT framework is composed of three main operational stages: data acquisition, data transmission, and data visualization & actuation. In the first stage, environmental parameters such as soil moisture, rainfall, temperature, and humidity are measured using sensors. These sensors are connected to a microcontroller unit (STM32F103C8T6) interfaced with a LoRa transceiver module (Ra-02 SX1278). This forms a sensor node in which the microcontroller processes sensor inputs and prepares data for transmission through LoRa. The use of LoRa solves internet connectivity problem of Bhuj where farms are 5 to 20 km from urban area. In the second stage, the processed sensor data is transmitted to a Collector node, which also comprises a LoRa transceiver(LoRa-1) and a more advanced microcontroller (STM32F407).

Collector node periodically initiates communication with each Sensor node, collects the sensor data at defined

intervals using time divisional multiplexing. The default operation on pump govern by existing logic on collector node. Users or automated rules can trigger commands (e.g., activating irrigation pumps) based on fuzzy logic conditions. The Collector node aggregates data and forwards it to the ESP32 node by using Radio-B(Frequency: 866.3000MHz). The ESP32 node is connected to the internet. The ESP32 node uploads data to Thingspeak server for visualization and monitoring purpose. The web server also enables bidirectional communication as the internet connection is available. The functionality of the sensor node is outlined in Algorithm 1. The sensor nodes employ a deep sleep cycle mechanism to optimize energy efficiency and extend battery lifespan. The Collector node is connected to two LoRa transceiver(LoRa-1 and LoRa-2). Collector node receives incoming sensor packets within their slots through Radio-A. It stores received data with timestamps. Collector After completion of cycle switch to Radio B for transmission of aggregated packets to ESP32 node. The working of Collector node is outlined in Algorithm 2. ESP32 node stays within the vicinity of internet connectivity. The main task of the ESP32 node is to receive data from the collector node via Radio-B. In this architecture all the field components deployed in the farm do not need internet connectivity. ESP32 node is the only node connected to the internet and sends data to the ThingSpeak cloud.

Algorithm 1. Sensor Node Operation(LoRa, 865.0625 MHz).

Require: Frequency f , Spreading Factor SF, Bandwidth BW, TX Power P_{tx} , Unique Node_ID; ACK timeout: T_{ack} , Cycle period: T_{cycle} , Max retries N

```
1: Initialise LoRa with ( $f$ , SF, BW,  $P_{tx}$ ); Set Node_ID
2: while true do
3:   data  $\leftarrow$  ReadSensors()
4:   pkt  $\leftarrow$  CreatePacket(Node_ID, Timestamp(), data)
5:   AppendCRC(pkt)
6:    $r \leftarrow 0$ , acked  $\leftarrow$  false
7:   repeat
8:     TX(pkt)
9:     ack  $\leftarrow$  WaitForACK( $T_{ack}$ )
10:    if ack = true then
11:      acked  $\leftarrow$  true
12:    else
13:       $r \leftarrow r + 1$ 
14:    end if
15:    until acked or  $r \geq N$ 
16:    if acked then
17:      Sleep( $T_{cycle}$ )
18:    else
19:      LogError("ACK not received after N retries")
20:      Sleep( $T_{cycle}$ )
21:    end if
22: end while
```

Require: Radio_A at 865.0625 MHz, Radio_B at 866.3000 MHz, ACK timeout $T_{ack,B}$; Max retries N

```

1: Init  Radio_A(865.0625 MHz),
Init    Radio_B(866.3000 MHz)
2: Init Data_Buffer  $\leftarrow \emptyset$ 
3: while true do
4:     Radio_A.listen()
5:     pkt  $\leftarrow$  Radio_A.receive()
6:     if pkt  $\neq \emptyset$  and CRC_OK(pkt) then
7:         Radio_A.send_ACK(pkt.source)
8:         Data_Buffer.push(Stamp(pkt))
9:     end if
10:    if Data_Buffer.not_empty() then
11:        item  $\leftarrow$  Data_Buffer.peek()
12:         $r \leftarrow 0$ , acked  $\leftarrow$  false
13:        repeat
14:            Radio_B.transmit(item)
15:            acked  $\leftarrow$  Radio_B.wait_ACK( $T_{ack,B}$ )
16:            if acked = false then
17:                 $r \leftarrow r + 1$ 
18:            end if
19:        until acked or  $r \geq N$ 
20:        if acked then
21:            Data_Buffer.pop()
22:        end if
23:    end if
24:    Radio_B.listen()
25:    cmd  $\leftarrow$  Radio_B.receive()
26:    if cmd  $\neq \emptyset$  then
27:        QueueForForward(cmd)
28:    end if
29: end while

```

3.3. IoT Communication and Cloud Integration (ThingSpeak)

The Thingspeak cloud analyzes data, and with fuzzy inference, it will make irrigation decisions. ThingSpeak is a free and open platform designed for the Internet of Things (IoT), facilitating the collection, analysis, and visualization of data derived from various sensors and devices. After being communicated to ThingSpeak, the data can be visualized on a dashboard, as depicted in Fig.

4. Users can receive alerts whenever certain environmental conditions exceed threshold. ThingSpeak dashboard also supports cloud storage for analysis of the data till current state. This improves decision making and allows for predictive irrigation strategies. The system ensures energy efficient, event driven data collection and remote actuation using LoRa based communication and cloud connectivity. The grouping of real time feedback, remote accessibility and cloud integration makes it a practical solution for precision agriculture and other similar applications.

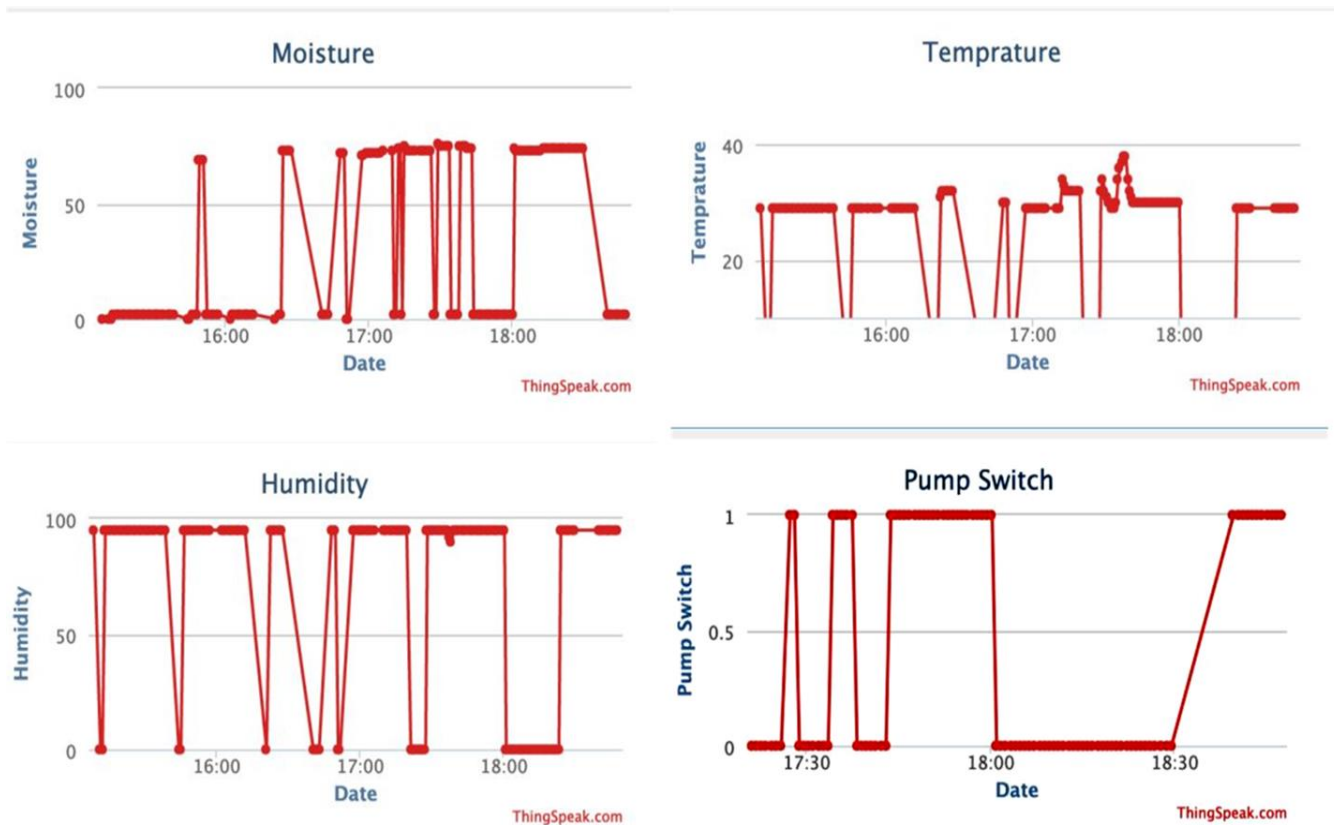


Fig. 4. Visualization of Data on ThingSpeak.

4. Fuzzy Inference System for Irrigation Pump Control

The fuzzy logic based system determines the proportion of pump openings based on the four input variables. Fuzzy logic based controllers provides number of advantages over traditional irrigation control systems particularly in situations with uncertainties and multi criterion decision-making [52,53]. Due to its dense and computationally effective representation, the proposed system use a Mamdani Fuzzy inference based system. The proposed system takes into account four presented input variables and utilizes fuzzy logic principles to determine the appropriate level of valve opening. The four input variables used in the proposed system are:

- The Soil Moisture Content: It indicates the current soil moisture level. It is important in determining watering timing for a specific plant. The fuzzy logic system receives moisture content measurements from the specified sensor.
- Ambient temperature: It impacts evaporation and plant water intake. Higher temperatures

accelerate evaporation and increase plant water needs.

- Relative Humidity: The ratio of atmospheric moisture to its maximal capacity at a given temperature. High humidity inhibits evaporation, influencing plant water needs.
- The Rain Water Level: It measures the quantity of rainfall received. After recent rains, soil moisture may be sufficient, decreasing the requirement for Irrigation.

The main advantage of fuzzy inference systems is the capability to effectively handle imprecise or ambiguous data, while also incorporating human skill and reasoning by utilizing "if-then" rules. The framework with the use of fuzzification, inference engine, and defuzzification helps the irrigation system to represent the nonlinear correlation between the input variables (soil moisture content, ambient temperature, relative humidity, rain water level) and the output (valve opening) in order to facilitate efficient irrigation control which is represented in Fig. 5.

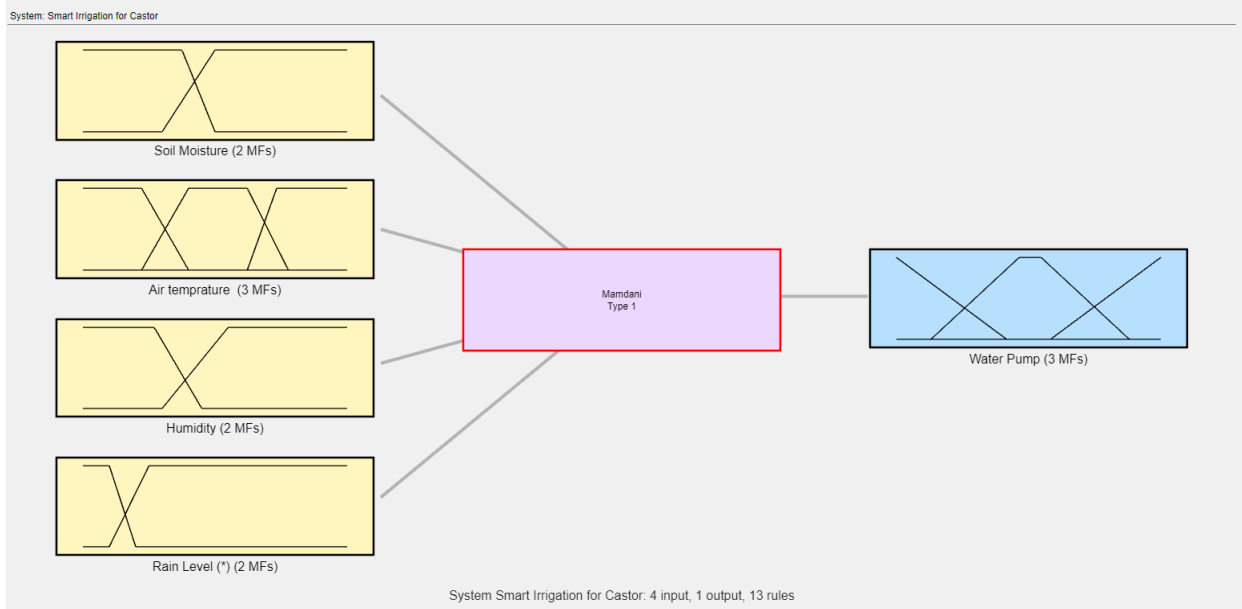


Fig. 5. Fuzzy Logic-Based Inference System for the proposed system.

4.1. Fuzzification

Here, every input value is assigned a degree of membership value inside one or more fuzzy sets. Membership functions are used to assign a numerical value varying from 0 to 1 to each input. The presented system employs trapezoidal, linear s-shaped saturation and Linear z-shaped saturation membership functions to transform the input parameters into fuzzy logic for irrigation scheduling. The trapezoidal membership

function is used for soil moisture representation, as it provides a simple yet effective transition between dry and moderate conditions. Additionally, the same membership function is suitable for temperature variations, as it accommodates a broader range of changes. The S-shaped membership function is used to parameters like humidity, where gradual transitions between low and high values occur, and the Z-shaped membership function is used for rainwater levels, ensuring a smooth decrease in irrigation demand as rainfall varies.

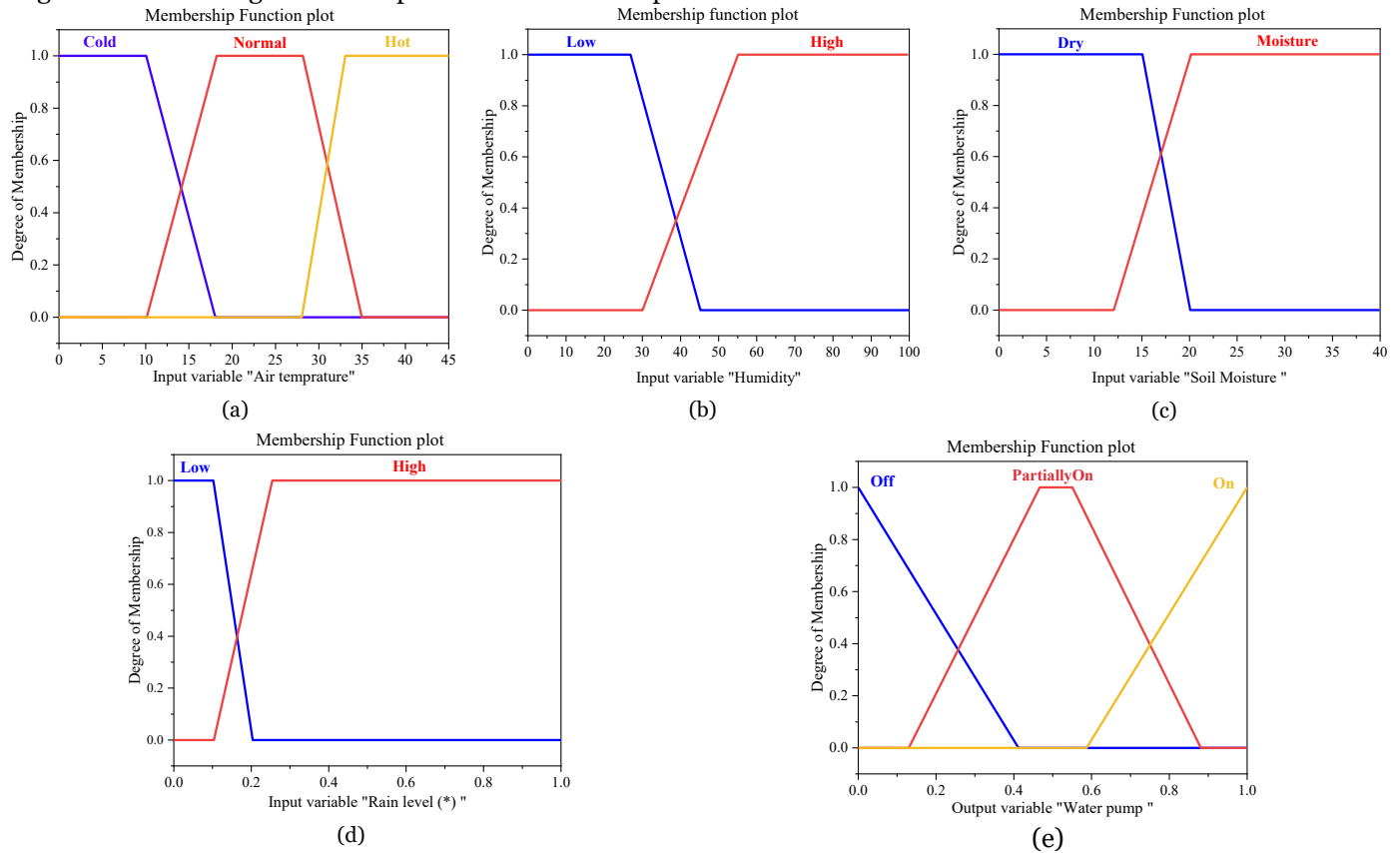


Fig. 6. Membership Functions for Input Parameters (Soil Moisture, Temperature, Humidity, Water Level) and Output (Valve Opening).

The trapezoidal membership function is described by four parameters as follows:

$$f(x, a, b, c, d) = \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right) \quad (4)$$

Where a is the lower limit, b is the lower support limit, c is the upper support limit and d is the upper limit.

Linear s-shaped saturation membership function is described by two parameters as follows:

- When $a < b$:

$$f(x, a, b) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases} \quad (5)$$

- When $a = b$:

$$f(x, a, b) = \begin{cases} 0, & x < a \\ 1, & x \geq a \end{cases} \quad (6)$$

Where "a" represents the foot, which signifies the initial point of the lower flat region. On the other hand, "b" denotes the shoulder, which signifies the point at which the curve begins to transition from the lower flat region to the upper flat region.

The membership function of linear z-shaped saturation is characterized by two parameters, which are as follows:

$$f(x, a, b) = \begin{cases} 1, & x < a \\ \frac{b-x}{b-a}, & a \leq x \leq b \\ 0, & x > b \end{cases} \quad (7)$$

- When $a = b$:

$$f(x, a, b) = \begin{cases} 1, & x < a \\ 0, & x \geq a \end{cases} \quad (8)$$

Where a defines the shoulder of the membership function, and b defines its foot.

The fuzzy membership functions of the proposed irrigation system are presented in the Fig. 6.

4.2. Inference Engine and Defuzzification

A fuzzy-based irrigation inference system revolves around the inference engine. It takes the input values that have been fuzzified and applies the fuzzy rules, which are usually "if-then" expressions. Whenever there are more than two input variables, these linguistic rules are logically connected using the AND, OR, or NOT connectives. With AND connectives connecting the inputs, Min-Max inference is used for the rule base in this situation. Table 2 shows the fuzzy rules that were formed for inference.

Defuzzification takes the fuzzy results from the inference engine and turns them into a clear, usable value. In most cases, the centroid method is applied, which finds the weighted average of the membership function. This value is then used to set the valve opening percentage based on sensor readings, making sure that water is supplied efficiently and according to the actual needs of the crop.

Table 2. Fuzzy Rules for Dynamic Valve Opening Proportion Calculation in MATLAB.

Sr. No.	Rule(s)	Weight	Name
1	If Soil Moisture is Dry and Air temprature is Cold and Humidity is Low and Rain Level (*) is Low then Water Pump is PartiallyOn	1	rule1
2	If Soil Moisture is Dry and Air temprature is Normal and Humidity is Low and Rain Level (*) is Low then Water Pump is On	1	rule2
3	If Soil Moisture is Dry and Air temprature is Hot and Humidity is Low and Rain Level (*) is Low then Water Pump is On	1	rule3
4	If Soil Moisture is Dry and Air temprature is Cold and Humidity is High and Rain Level (*) is Low then Water Pump is PartiallyOn	1	rule4
5	If Soil Moisture is Dry and Air temprature is Normal and Humidity is High and Rain Level (*) is Low then Water Pump is On	1	rule5
6	If Soil Moisture is Dry and Air temprature is Hot and Humidity is High and Rain Level (*) is Low then Water Pump is On	1	rule6
7	If Soil Moisture is Moisture and Air temprature is Cold and Humidity is Low and Rain Level (*) is Low then Water Pump is Off	1	rule7
8	If Soil Moisture is Moisture and Air temprature is Normal and Humidity is Low and Rain Level (*) is Low then Water Pump is PartiallyOn	1	rule8
9	If Soil Moisture is Moisture and Air temprature is Hot and Humidity is Low and Rain Level (*) is Low then Water Pump is On	1	rule9
10	If Soil Moisture is Moisture and Air temprature is Cold and Humidity is High and Rain Level (*) is Low then Water Pump is Off	1	rule10
11	If Soil Moisture is Moisture and Air temprature is Normal and Humidity is High and Rain Level (*) is Low then Water Pump is Off	1	rule11
12	If Soil Moisture is Moisture and Air temprature is Hot and Humidity is High and Rain Level (*) is Low then Water Pump is PartiallyOn	1	rule12
13	If Soil Moisture is Dry and Rain Level (*) is High then Water Pump is Off	1	rule13

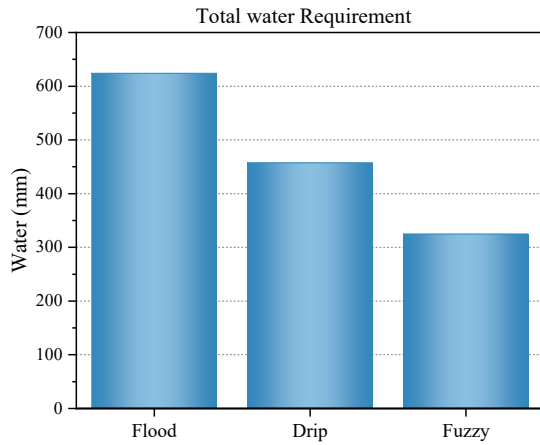
5. Results and Discussion

The proposed fuzzy-IoT irrigation system was deployed in a castor field located near Bhuj, India, to evaluate its effectiveness under real-world conditions. The experiments were conducted using the GC-2 variety of castor. A comparative assessment of different irrigation approaches such as flood irrigation, smart drip irrigation, and the proposed fuzzy IoT system was performed with a specific focus on the total water provided, expressed in millimeters (mm) as presented in Table 3 and Fig. 7. The proposed system had the lowest total water requirement at 335.3 mm, next was drip irrigation (466mm) [54,55], and a very large difference to flood Irrigation (635 mm). When expressed in volumetric terms, the proposed system minimized water use to 337.91 m³, reducing consumption by 47.18% compared to flood irrigation and 26.61% compared to drip irrigation. These findings are consistent with previous studies [55-57], which report average water savings of 25-30% through drip based methods.

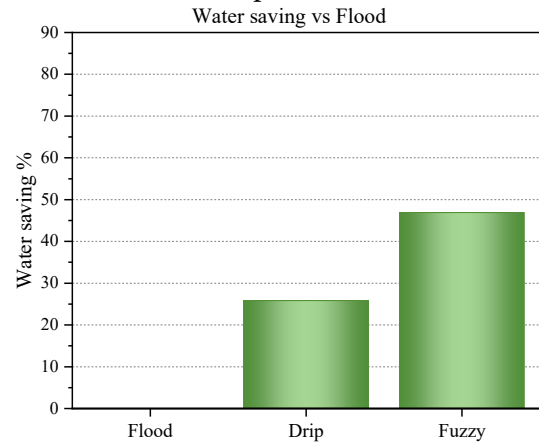
Table 4 and Fig. 8 provides a comparative overview of recent fuzzy logic based irrigation systems, highlighting their technology stack, control logic, water saving potential, and application contexts. A key

distinction of the proposed Fuzzy IoT system is that it represents one of the few real field deployments. For example, [58] reported a continuous-state fuzzy controller tested in a laboratory environment, achieving a water saving of 23.1%. Similarly, [53] demonstrated a rule-based fuzzy system in a small orchard in Morocco, with water savings of 18–21%. However, [59] and [60], presented simulated or prototype scale systems integrating edge intelligence and optimization, but without open-field validation. Even where field experiments were conducted, such as in the study by [45] on litchi orchards, deployment was confined to subtropical environments.

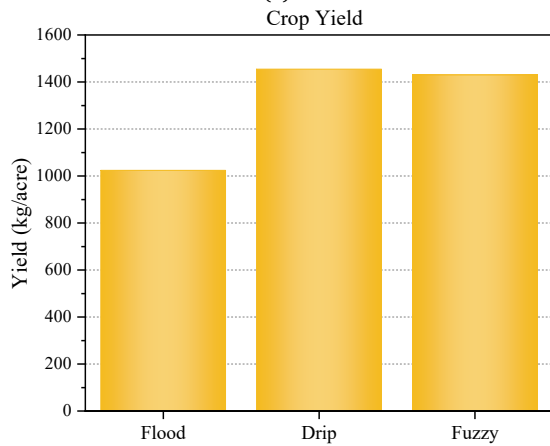
The proposed system demonstrates a water saving of 26%, outperforming the efficiency reported in most comparable studies. An additional advantage of the proposed framework is that it does not require continuous internet connectivity at the farm site, as the decision-making and control processes are executed locally. This makes the system particularly well-suited for deployment in remote or rural regions, where internet availability is often unreliable or absent. The reported for water saving percentages across different studies are not directly comparable because deployment conditions are different. The values should be interpreted as indicative trends rather than strict performance evaluations.



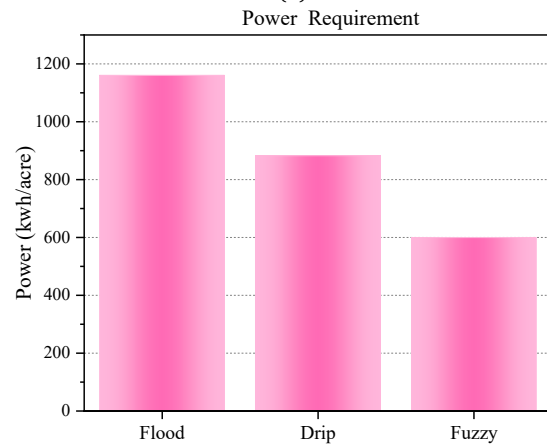
(a)



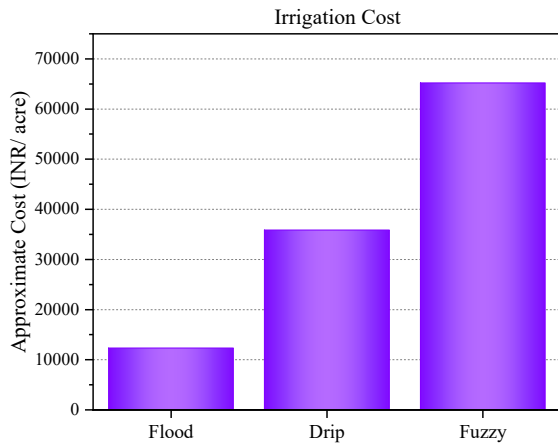
(b)



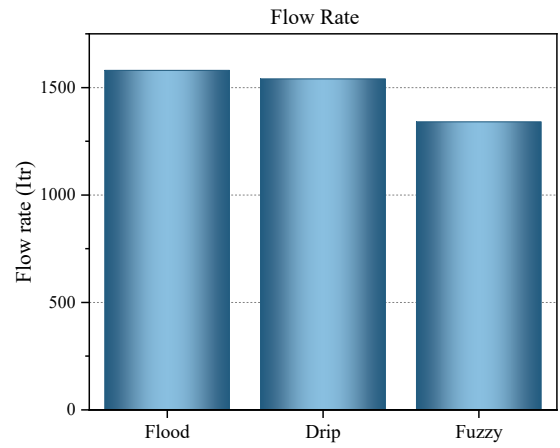
(c)



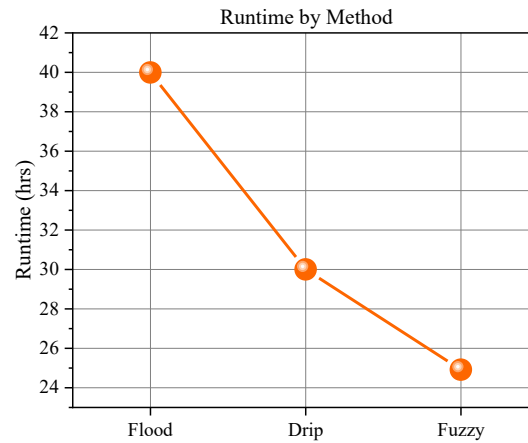
(d)



(e)



(f)



(g)

Fig. 7. Comparison of Fuzzy IoT with conventional irrigation methods highlighting water usage, savings, yield, power consumption, and efficiency.

Several factors determine the power consumption of Internet of Things (IoT) smart irrigation systems. These include system size, sensor and actuator complexity, data transmission and analysis frequency, and the number of actuators and sensors used. Table 5 presents a concise

overview of power consumption for the hardware components mentioned, measured in milliamperes (mA) and milliwatts (mW), under the assumption of a supply voltage of 5V of the proposed system.

Table 3. Comparative assessment of different irrigation approaches.

Evaluation Criteria	ETc-Based	Flood Irrigation	Drip Irrigation	Fuzzy-IoT Irrigation
Total Water Requirement (mm)	400	635	466	335.3
Water Requirement (m ³ /acre)	–	639.97	469.63	337.91
% Water Saving (vs Flood)	37.01%	–	26.61%	47.18%
Crop Yield (kg/acre)	–	1032	1470	1445.9
Irrigation Cost (INR/acre)	–	Rs 12,550	Rs36,100	Rs65,000
Motor Rating (HP / kW)	–	4.0 HP / 2.98 kW	4.0 HP / 2.98 kW	3.5 HP / 2.61 kW
Average Flow Rate (L/hr)	–	~1600	~1560	~1350
Average Daily Runtime (hrs)	–	~4.0	~3.0	~2.5
System Pressure Required	–	Low (0.5-1 bar)	Medium (1.5-2 bar)	Medium-High (2-3 bar)

Table 4. Comparison with recent fuzzy logic-based irrigation systems.

Study	Technology	Control Logic	Water Saving (%)	Application Site
[58]	ESP32, LoRa, Cloud Fuzzy Controller	Continuous-state fuzzy logic	23.1%	Laboratory-scale HIL setup
[53]	Arduino, LoRa, Environmental Sensors	Rule-based fuzzy control	18-21%	Small orchard, Morocco
[59]	Raspberry Pi, Soil & Camera Sensors	Fuzzy + Edge ML integration	Not evaluated	Simulation
[60]	NodeMCU, DHT11, Soil Moisture Sensors	Optimized fuzzy (GWO/GA)	20-24%	Lab-scale prototype

Study	Technology	Control Logic	Water Saving (%)	Application Site
[45]	Fuzzy Controller + Humidity & Temperature Sensors	Multi-layer fuzzy logic	~20%	Field, subtropical orchard (Litchi)
[61]	Fuzzy Controller	Temperature, soil moisture	ET, 13%	Modeled turf/grass; Turkey climate data
Fuzzy-IoT(Bhuj, India, 2025)	STM32, LoRa, Soil/Moisture/Temperature Sensors	Fuzzy logic	26%	Real field deployment, 1 acre, GC-2 castor

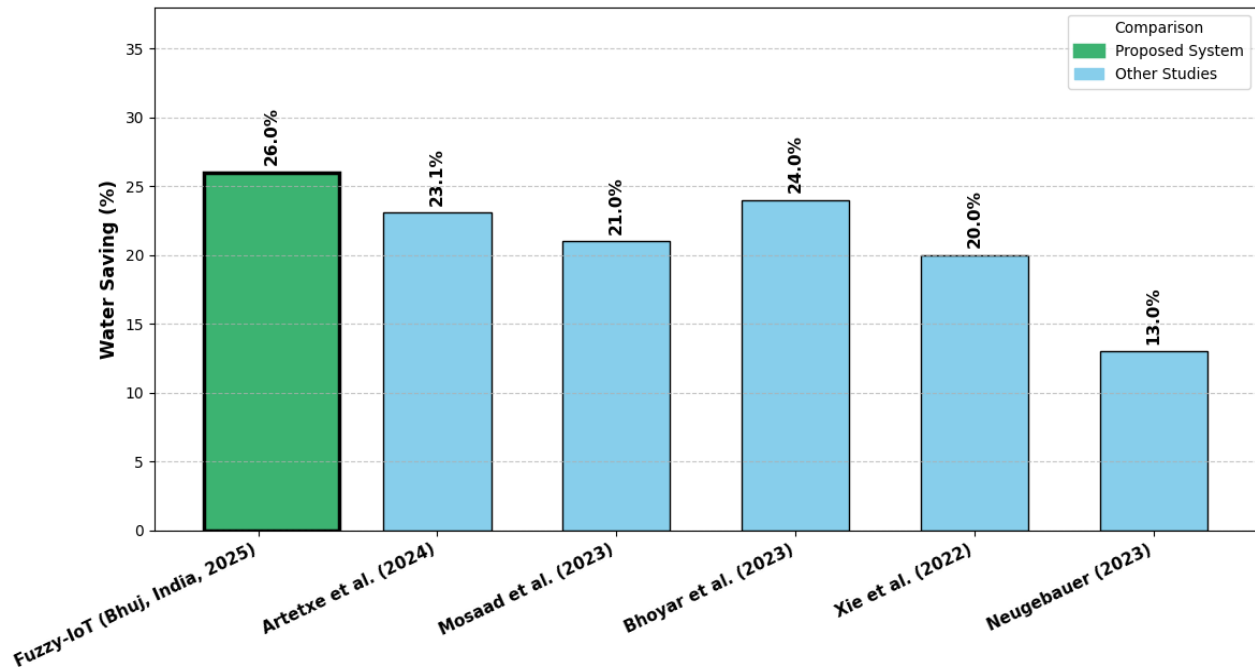


Fig. 8. Comparison with State of art Fuzzy Irrigation Systems.

Table 5. Power consumption by different hardware components.

Hardware Components	Quantity	Operating Current (mA)
Soil moisture sensor(YL69)	1	10
DHT11 Digital Relative Humidity and Temperature Sensor	2	$2.2 \times 2 = 4.4$
Rainwater level sensor	1	$2 \times 15 = 30$
STM32F407VG Collector Node	1	12.7
STM32F103C8T6 Sensor Node	3	$8.4 \times 3 = 25.2$
Lora Antenna's 865 Mhz Wi-Fi Transmitter Module	6	$11 \times 6 = 66$
ESP32:	1	40
VFD Module	1	5
Total Current Consumer (mA)		141.3 mA
Supply Voltage		5V
Power Consumption(in Watts)		706 mW

5.1.Future Direction

The future work will focus on statistical analysis and provides error graph with conventional techniques over a period of time. It will also integrate the predictive analysis on the real time data and past stored data to provide the irrigation schedule also for better decision making. It has also good scope of work to combine fuzzy logic with machine learning models to give best water distribution as per varying field conditions. Field trails

will be conducted in diverse climatic conditions for the performance analysis across different soil types. Another key scope of the work is power consumption and time span estimation to provide power from battery. There is also a way to provide a power for the charging battery enable system though solar panels. For real time farming applications, cost analysis, economical feasibility, operational cost can be compared with conventional techniques in context to improvement in yield and return on investment to farmers.

6. Conclusion

This work designed and implemented a Fuzzy IoT irrigation system using the GC-2 variety of castor in Bhuj, India. The system integrated multiple sensors, including humidity, rainfall, soil moisture, and temperature, to optimize irrigation management using fuzzy-based decision-making. The motor pump valve is operated according to the decision. The proposed system operated a motor pump without requiring internet infrastructure via LoRa-based wireless communication. The system eliminated the need for continuous internet connectivity at the farm. The system integrated multiple sensors, including humidity, rainfall, soil moisture, and temperature, to optimize irrigation management using fuzzy-based decision-making. The system is implemented using an STM32 Microcontroller, an ESP32 microcontroller, and LoRa for long-range wireless communication. It offered infrastructure-less connectivity for long-range communication and reliable operation without internet connectivity in the field. The proposed system addressed water scarcity challenges more effectively than existing fuzzy IoT approaches and traditional irrigation methods, leveraging long-range wireless communication via LoRa and low-power electronic devices. This work presented the design and deployment of a Fuzzy-IoT irrigation system specifically tailored for castor cultivation, validated under real field conditions in Bhuj, India, using the GC-2 variety. A fuzzy inference system estimates weather conditions (temperature, relative humidity), rainfall level, and soil moisture to improve the accuracy of irrigation decisions, particularly valve actuation for the motor pump. By integrating fuzzy control with LoRa-based wireless sensor networks, the proposed system addressed critical challenges in irrigation scheduling, data transmission, and automated pump actuation. Importantly, the architecture eliminates the need for continuous internet connectivity at the farm, enabling reliable operation in rural and connectivity-constrained regions. The proposed fuzzy-IoT system achieved the lowest water requirement (337.91 m³/acre), delivering 26% savings compared to drip irrigation and 47% compared to flood irrigation. The performance is better than that of prior irrigation systems and has been uniquely validated in real field castor cultivation. Overall, the study demonstrates that combining fuzzy logic with IoT sensing and LoRa communication enables precise, crop-specific, and internet-independent irrigation management.

Conflicts of Interests

There is no conflict of interest in this study.

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Availability of Data and Materials

The datasets used during the current study are available from the corresponding author on request.

Using Artificial Intelligence Chatbots

None.

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